scientific reports

OPEN

Machine learning-based integration develops relapse related signature for predicting prognosis and indicating immune microenvironment infiltration in breast cancer

Junyi Li[1,3], Shixin Li[2,3], Dongpo Zhang[1,3], Yibing Zhu[1,3], Yue Wang[1], Xiaoxiao Xing[1], Juefei Mo[1], Yong Zhang[1], Daixiang Liao[1]✉ & Jun Li[1]✉

Breast cancer is the most common type of cancer in women, and while current treatments can cure the majority of early-stage primary BC cases, recurrence remains a significant challenge. Traditional methods of assessing patient prognosis, such as AJCC, TNM staging, and biochemical markers, are no longer sufficient in the era of precision medicine. Existing tumor models often rely on single selection and simpler algorithms, which can lead to poor effectiveness or overfitting. To address these limitations, this study systematically analyzed RNA-seq high-throughput data and combined 10 machine learning algorithms to construct 117 models. The optimal algorithm combination, StepCox[both] and ridge regression, was identified, and an immune-related gene signature (IRGS) composed of 12 genes was developed. The IRGS demonstrated outstanding predictive performance across multiple datasets and surpassed 10 previously published signatures. GSEA analysis revealed significant enrichment differences in cellular processes, diseases, and immune-related pathways between high- and low-risk recurrence patients. The low recurrence risk group based on IRGS exhibited a stronger immune phenotype and better survival prognosis, which may be associated with higher infiltration of CD4 + and CD8 +T cells. However, high M2 macrophage infiltration suggests potential immune escape in low recurrence risk patients. Combined with immune checkpoint expression levels and TIDE results, it is suggested that low-risk patients may respond positively to immunotherapy. Through drug sensitivity analysis, potential drugs that are more effective for both high- and low-risk groups have been identified. Therefore, the IRGS developed in this study can serve as an adjunct tool for assessing the recurrence risk of breast cancer, potentially enhancing personalized treatment planning, and improving the clinical management of patients with breast cancer.

Keywords Data mining, Prognosis, Immune microenvironment, Expression difference, Functional enrichment analysis, Machine learning

In 2020, statistics from 110 countries revealed that breast cancer has the highest incidence of cancer worldwide[1]. Breast, lung, and colorectal cancers collectively represent 50 percent of all cancer diagnoses in female patients[2]. Possibly due to the decrease of human fertility and weight gain, breast cancer alone accounts for 30% of diagnosed cancers in women, with its prevalence continuing to rise annually[3].

In recent years, with the continuous development of gene sequencing and biomedical technology, breast cancer typing has become increasingly refined[4]. As a result, treatment methods have become more abundant, leading to significant progress in the treatment of breast cancer. Depending on the molecular subtype, chemotherapy,

[1]Department of Surgery, Guang 'anmen Hospital, China Academy of Chinese Medical Sciences, Beijing 100053, People's Republic of China. [2]Department of Oncology, The First Affiliated Hospital of Guizhou University of Traditional Chinese Medicine, Guiyang 550025, Guizhou, People's Republic of China. [3]Junyi Li, Shixin Li, Dongpo Zhang and Yibing Zhu contributed equally. ✉email: gamyyljy@163.com; jli.doctor1972@icloud.com

endocrine therapy, ERBB2-targeted intervention, or a combination thereof may be administered reasonably[5]. Additionally, new drug therapies such as anti-programmed cell death ligand 1 (PD-L1) immunotherapy, CDK4/6 inhibitors (cyclin-dependent kinase 4/6), and phosphatidylinositol 3 kinase (PI3 K) inhibitors are also available[6–8].

Current treatments can cure most early primary breast cancer cases. However, metastasis and recurrence remain the leading causes of death related to breast cancer. Risk factors for relapse may include subclinical infections, smoking, chronic inflammation, use of foreign estrogens, alcohol abuse, major depression, and low economic conditions[9–11]. Resistance in patients with ER-positive breast cancer treated with tamoxifen and aromatase inhibitors is a result of limited understanding of recurrence mechanisms and the absence of preclinical models. This resistance can lead to local recurrent or metastatic breast cancer during or after adjuvant endocrine therapy[12]. Study showed that tumors characterized as ER-, HER2 +, and PR- exhibit a higher risk of recurrence. Specifically, the 5-year recurrence rate for ER- breast cancer is significantly higher compared to ER + tumors[13,14]. The recurrence time of 2,730 breast cancer patients was analyzed to investigate the relationship between molecular subtypes and overall survival (OS). The findings revealed that early recurrence (within 5 years of diagnosis) is a critical prognostic factor for OS across all subtypes. In contrast, late recurrence (after 5 years of diagnosis) is only a significant prognostic factor for OS in the Luminal B subtype, with a hazard ratio (HR) of 4.30[15]. Additionally, data from 2,992 patients revealed that ER negativity is associated with early recurrence but not late recurrence. Furthermore, their study highlighted that Asian, Black, and Hispanic women have a higher risk of early recurrence compared to non-Hispanic White women[16]. While increased public awareness has improved early detection and treatment of breast cancer, many patients tend to overestimate the risk of recurrence after treatment. This presents challenges as overestimating the perceived risk of recurrence and associated fear can have negative impacts on long-term outcomes for cancer survivors[9]. Therefore, there is an urgent need for better techniques to identify which patients are at risk of relapse.

In this study, our focus was on the recurrence of breast cancer. We utilized expression data and patient information from 3,620 breast cancer patients across 6 independent public datasets. After analyzing key pathways related to relapse using Gene Set Enrichment Analysis (GSEA), we developed a risk signature based on key pathway genes and validated it using machine learning methods. The predictive effect was evaluated in external datasets, and its validity and superiority were compared with ten published signatures. Furthermore, we analyzed the relationship between the risk signatures and the immune microenvironment of patients, as well as their responses to immune checkpoint inhibitor (ICI) therapy and targeted drugs. This work has the potential to optimize precision therapy and improve clinical outcomes for breast cancer patients (Fig. 1).

Materials and methods

High-throughput sequencing data collection and processing

The high-throughput sequenced gene expression data and clinical information for the breast cancer cohort were obtained from the Consortium International Molecular Classification of Breast Cancer (METABRIC), the NCBI Integrated Gene Expression Database (GEO), and the Cancer Genome Atlas (TCGA) database (Table S1). The METABRIC cohort, serving as a training cohort, comprised 1980 patients with breast cancer, which was used to construct prognostic signature. The TCGA-BRCA dataset included 1102 breast cancer tissues and 113 normal tissues. Additionally, datasets GSE20685, GSE20711, GSE25066, and GSE42568 were downloaded from the NCBI GEO database (https://www.ncbi.nlm.nih.gov/geo/). Among these datasets, GSE20685, GSE20711, GSE25066, and GSE42568 contained tumor tissue samples from 327, 90, 170, and 121 patients respectively. These samples were sequenced based on the GPL570 [HG-U133_Plus_2] Affymetrix Human Genome U133 Plus 2.0 Array. Another GEO dataset, GSE25066 contained tumor tissue samples from 508 patients with sequencing information based on the GPL96 [HG-U133A] Affymetrix Human Genome U133A Array. The METABRIC, GSE20711 and GSE42568 cohorts were utilized for the GSEA analysis and signature construction through machine learning methods. The GSE25066, GSE20711, and TCGA datasets were employed as external cohorts to assess the validity and accuracy of prognostic signature. The RNA-Seq expression levels generated by the workflow were read and normalized using the quartile of transcription expression value FPKM (FPKM-UQ). The differential gene expression between breast cancer tissue and normal tissue was determined based on the R package limma with voom method[17]. A threshold value of false discovery rate (FDR) <0.05 and relative expression value (|log2 fold change (FC) |) $>=1.0$ was applied to identify significant gene differences. The results of gene differential analysis were visualized using ggplot2 R package.

Based on relapse-free survival (RFS) data, patients who relapsed within five years (RFS.event = 1 or 0, RFS.time < 5 years) were classified into the short-term group (ST), and patients who were relapse-free for more than five years (RFS.event = 0, RFS.time > 5 years) were categorized as the long-term group (LT). Accordingly, the METABRIC cohort contains 1342 LT samples and 495 ST samples, the GSE20711 contains 51 LT samples and 31 ST samples, the GSE42568 contains 54 LT samples and 41 ST samples, the GSE25066 contains 61 LT samples and 110 ST samples, and the GSE20685 contains 238 LT samples and 15 ST samples. The TCGA dataset included 209 breast cancer patients in the LT group and 86 patients in the ST group.

GSEA

The complete symbolic gene collection of the Kyoto Encyclopedia of Genes and Genomes is accessible through the GSEA database (http://www.gsea-msigdb.org/gsea/). Using the SNR measurement in GSEA4.3.3 software, the normalized expression data of LT and ST samples in the above dataset were uploaded. Based on the Kyoto Encyclopedia of Genes and Genomes pathway (KEGG, c2.cp.kegg.7.5.1.symbols)[18], the correlation between gene expression and RFS was analyzed. Gene sets containing fewer than five genes, or more than 2,000 genes were excluded from the analysis. Normalized enrichment score (NES) |> 1, $p<0.05$, FDR <0.25 were utilized as cut-off values for evaluating enrichment and statistical significance.

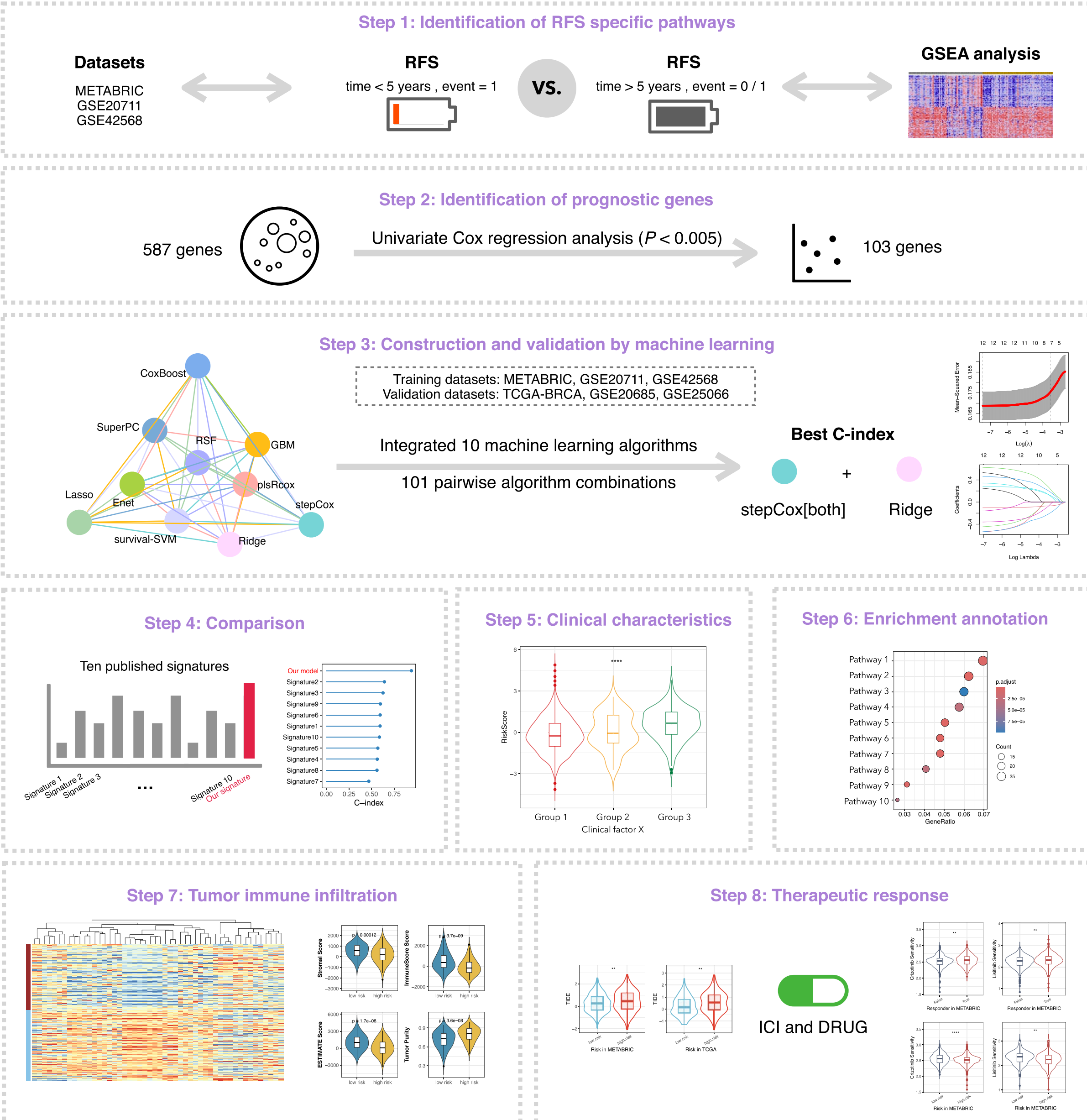


Fig. 1. The workflow in this study.

Prognostic signature generated by an integrated approach based on machine learning

To develop a recurrence prognosis signature with high accuracy and stability, we integrated 117 algorithmic combinations of 10 machine learning algorithms based on the mlr3 R package (https://seandavi.github.io/ITR/machinelearningmlr3.html). The comprehensive algorithms include random survival forest (RSF), elastic network (Enet), minimum absolute contraction and selection operator (Lasso), Ridge, step by step Cox (stepCox), Cox boost, Cox's Partial least squares regression (plsRcox), supervised Principal component (SuperPC), generalized Enhanced regression modeling (GBM), and survival support vector Machine (Survival-SVM)[19–27]. First, univariate Cox regression was utilized to identify relapse-specific pathway genes associated with prognosis from GSEA analysis for the METABRIC, GSE20711 and GSE42568 cohorts. Subsequently, a combination of 117 algorithms was employed, based on the Leave-one-out cross-validation (LOOCV) framework, to construct a prediction model using these prognostic genes in the METABRIC, GSE20711 and GSE42568 cohorts. To optimize the model selection process, we systematically explored all possible combinations of directional parameters, including "both," "backward," and "forward." For each algorithm combination, the Harrell's concordance index (C-index) was calculated to evaluate the model's predictive performance using rms package. The C-index serves as a robust metric for assessing the concordance between predicted and actual outcomes, with higher values indicating better model performance. After evaluating all algorithm combinations, the combination yielding the highest average C-index across the validation sets was selected as the optimal model. Furthermore, the predictive validity of signatures was assessed on the GSE25066, GSE20685 and TCGA datasets. As a result, a polygenic prognostic signature based on RFS time of breast cancer patients was established, and the risk score (RS) of patients was calculated according to a specific formula.

$$Risk\ score = \sum\nolimits_{i=1}^{n} Coef_i * x_i$$

Breast cancer patients were divided into high-risk and low-risk groups based on the RS median value. Kaplan–Meier method and Log-rank test were used to analyze the survival rate of patients. The timeROC R package was used to calculate the area under Receiver Operating characteristic (ROC) curve (AUC) of patients in the training set and validation set to evaluate the validity and sensitivity of prognostic signature. According to the information of lymph node metastasis, clinicopathology, histological stage and age, patients were divided into different subgroups, and the relationship between prognostic signature and clinical indicators was analyzed.

Survival analysis

To investigate the relationship between the expression of prognostic signature genes and OS or RFS in patients, the expression profile was normalized to logarithm. Univariate Cox regression analysis was conducted ($p < 0.05$), and HR and 95% confidence interval (CIs) were calculated. Using the median value of expression level as the threshold, tumor samples were categorized into high/low expression level groups. Based on R package Survival, Kaplan–Meier method and Log-rank test were employed to assess and illustrate the survival curve of patients based on gene expression. Genes significantly associated with patient survival (Log-rank test, $p < 0.05$) were identified as prognostic genes.

Performance comparison of prognostic signatures

We conducted a comprehensive literature search to identify 10 reported prognostic signatures for breast cancer, including Signature 1 (a ferroptosis-related prognostic signature)[28], Signature 2 (a pyroptosis-related prognostic signature)[29], Signature 3 (a cuproptosis/ferroptosis-related gene signature)[30], Signature 4 (a cancer-associated fibroblast signature)[31], Signature 5 (a degradome-based prognostic signature)[32], Signature 6 (a cuproptosis-related signature)[30], Signature 7 (a prognostic 6-gene signature for breast cancer based on multi-omics and single-cell data)[33], Signature 8 (a glycolysis-related four-mRNA signature)[34], Signature 9 (an immune-related signature)[35] and Signature 10 (a pyroptosis-related signature)[36] (Table S2). We calculated the HR, AUC, and p values through Log-rank test analysis. Subsequently, we compared the predictive performance of our prognostic signatures with the 10 reported prognostic signatures to determine whether machine learning-based prognostic signatures demonstrate superior performance.

Gene ontology and KEGG functional enrichment analysis

Based on the GO and KEGG, differential genes between high and low risk groups were analyzed using the R packages ClusterProfile and ggplot2. The potential biological functions were enriched and examined in terms of cell composition (CC), molecular function (MF), biological process (BP), biological pathways, diseases, and drugs[37].

Assessment of the tumor microenvironment

Based on RNA-seq data from the METABRIC cohort, we utilized the ESTIMATE algorithm to compute the ESTIMATE score, tumor purity, stromal and immune scores in order to characterize the TME[38]. The stromal and immune scores indicate the levels of infiltration of stromal cells and immune cells in tumor tissue, respectively. The ESTIMATE score was integrated with the stromal score and immune score to assess TME along with the tumor purity score[39]. A higher stromal and immune score corresponds to a lower tumor purity score, indicating that increased infiltration of stromal and immune cells in tumor tissue is associated with decreased tumor purity. Violin charts were employed to visualize differences in microenvironment scores between high- and low-risk groups as well as among each clinical factor subgroup.

Expression abundance of tumor-infiltrating immune cells (TIICs)

We assessed the abundance of 22 TIICs in all samples using 5 tumor immune algorithms of the R package IOBR, including TIMER[40], CIBERSORT[41], QUANTISEQ[42], MCPcounter[43], and EPIC. The correlation between prognostic features and immune cell infiltration was examined. Heat maps were utilized to display correlation matrices for risk score, prognostic signature gene, tumor purity, immune score, stromal score, ESTIMATE score, and TIICs abundance. Additionally, a box chart was employed to illustrate the variance in TIICs abundance between high-risk and low-risk groups calculated by CIBERSORT.

Evaluation of immunotherapy response

The expression levels of common immune checkpoints were compared between high and low risk groups. The Tumor Immune Dysfunction and Exclusion Algorithm (TIDE, http://tide.dfci.harvard.edu/) was utilized to analyze the immunotherapy biomarkers between high-risk and low-risk groups in order to assess the breast cancer prognosis signature for predicting the efficacy of ICI effect[44]. A high TIDE score was associated with no response to immunotherapy, while a low TIDE score indicated the opposite outcome.

Evaluation of chemotherapy response

According to Genomics of Drug Sensitivity in Cancer (GDSC, https://www.cancerrxgene.org/) data, prediction and analysis of patient's response to chemotherapy or targeted drugs, The R package pRophetic was then used to compare a semi-inhibitory concentration (IC50 value) to assess drug sensitivity in high-risk and low-risk patients[45].

Statistical analysis

All data processing, statistical analysis, and plotting were conducted in R 4.4.1 software. The Shapiro–Wilk and Kolmogorov–Smirnov tests were conducted to assess the data normality. For data that did not meet the normality assumption, we applied the Wilcoxon rank-sum test for differential analysis. We also performed a log-transformation on the data and re-assessed normality, which resulted in a *p*-value greater than 0.05, confirming a normal distribution. Then the effect sizes and fold changes (FC) were calculated for the different groups. A FC greater than 1.5 coupled with a *p*-value less than 0.05 was considered indicative of significant differences between groups. Statistical significance of two groups of variables fitted to a normal distribution was estimated by unpaired Students' *t*-test. While for data that did not meet the normality assumption, we applied the Wilcoxon rank-sum test for differential analysis. To perform multiple group comparisons, Analysis of Variance (ANOVA) followed by Tukey's Honestly Significant Difference (HSD) test was performed to evaluate differences among all components. Spearman's correlation coefficient was used for correlation analysis between two groups of variables.

Results

Recurrence-related pathways in BRCA

To explore the key biological pathways linked to tumor recurrence in breast cancer, patients were categorized into LT and ST groups based on RFS clinical data. Using the GSEA method based on the c2.cp.kegg.7.5.1.symbols gene set, pathway enrichment analyses were performed for both groups. As shown in Table S3, the ST group exhibited significant enrichment in cell growth/death, transport/catabolism, cell community, and immune system pathways (NES < -1, $p < 0.05$, FDR < 0.25). Conversely, the LT group showed significant enrichment in signal transduction, genetic information processing, metabolism, and endocrine system pathways (NES > 1, $p < 0.05$, FDR < 0.25). The enrichment results revealed that some gene sets were co-enriched across different datasets. For instance, the GSE20711 and GSE42568 cohorts were co-enriched in the chemokine (hsa04062) and B cell receptor (hsa04662) signaling pathways. Additionally, the GSE20711 and METABRIC cohorts displayed co-enrichment in the Toll-like receptor signaling pathway (hsa04620). However, there were also instances where gene sets were enriched by a single dataset, likely due to sample heterogeneity across different datasets (Fig. 2A). Despite this, the classification of pathway subgroups in different datasets demonstrated a consistent enrichment trend. Notably, the ST group significantly enriched a wide range of immunomodulatory-related pathways, including the chemokine signaling pathway (hsa04062), B cell and T cell receptor signaling pathway (hsa04662, hsa04660), NOD-like receptor signaling pathway (hsa04621), and natural killer cell-mediated cytotoxicity (hsa04650) (Fig. 2B). These findings indicate that immune-related pathways play a crucial role in the ST group and are likely to be closely associated with cancer recurrence.

Machine learning-based construction of the immune-related gene signature

An immune-related gene signature (IRGS) was developed through a machine learning-based integrative approach, leveraging the expression profiles of 587 major genes in the immune-related pathways. Univariate Cox analysis ($p < 0.005$) identified 103 prognostic genes, which were used in construct prediction models via a machine learning integration program. A total of 117 models were developed using the LOOCV framework, with the C-index calculated for each model across METABRIC, GSE42568 and GSE20711 cohorts. The optimal model, combining StepCox[both] and Ridge, achieved the highest average C-index (0.84) and consistently high C-index values across all datasets (0.823, Fig. 3A). This model yield 55 genes through StepCox[both] proportional risk regression followed by Ridge regression to obtain the optimal λ when the partial likelihood deviance reached the minimum value and identified the final set of 12 genes, including CHUK, BLNK, VAV3, MAP2K4, FOS, GZMB, TANK, SERPINA5, MS4A1, TXN, CD1C, and CDC42 ($p < 0.05$; Fig. 3B, 3C; Table S4).

Evaluation of the IRGS signature

Subsequently, the predictive performance of IRGS was assessed using ROC analysis. The 1-year, 3-year, and 5-year AUC values for METABRIC were 0.769, 0.872, and 0.877, respectively. For GSE42685, the AUC values 0.739, 0.755, and 0.762, while for GSE20711, they were 0.742, 0.797, and 0.866 (Fig. 3D). Collectively, these findings demonstrated the stable and reliable performance of IRGS across multiple independent cohorts. Then, individualized risk scores for patients were calculated by weighting the expression levels of IRGS genes using regression coefficient in a risk-scoring formula. Patients were divided into high- and low-risk groups based on the median RS. As shown in Fig. 3E and 3F, the RFS in the high-risk group was significantly lower than that in the low-risk group, both in the training and external validated cohorts (HR > 1, $p < 0.05$).

Comparison of prognostic signatures in breast cancer

Advanced in second-generation sequencing and big data technologies have driven the development of numerous prognostic polygenic markers based on risk regression and machine learning. To evaluate the superiority of our IRGS, we conducted a comprehensive review of published signatures, selecting 10 signatures and 82 features associated with various biological processes, including immune response, autophagy, iron death, desiccation, epithelial-mesenchymal transition, toll-like receptor signaling, etc. Our analysis revealed that IRGS demonstrates superior stability and predictive performance across datasets. Univariate Cox regression showed that IRGS effectively determined patient survival in every cohort, a stability not matched by other models of IRGS (Fig. 4A). Furthermore, IRGS consistently achieved a higher C-index compared to other models (Fig. 4B, 4 C), indicating its superior predictive accuracy. Unlike many other signatures that perform well on datasets

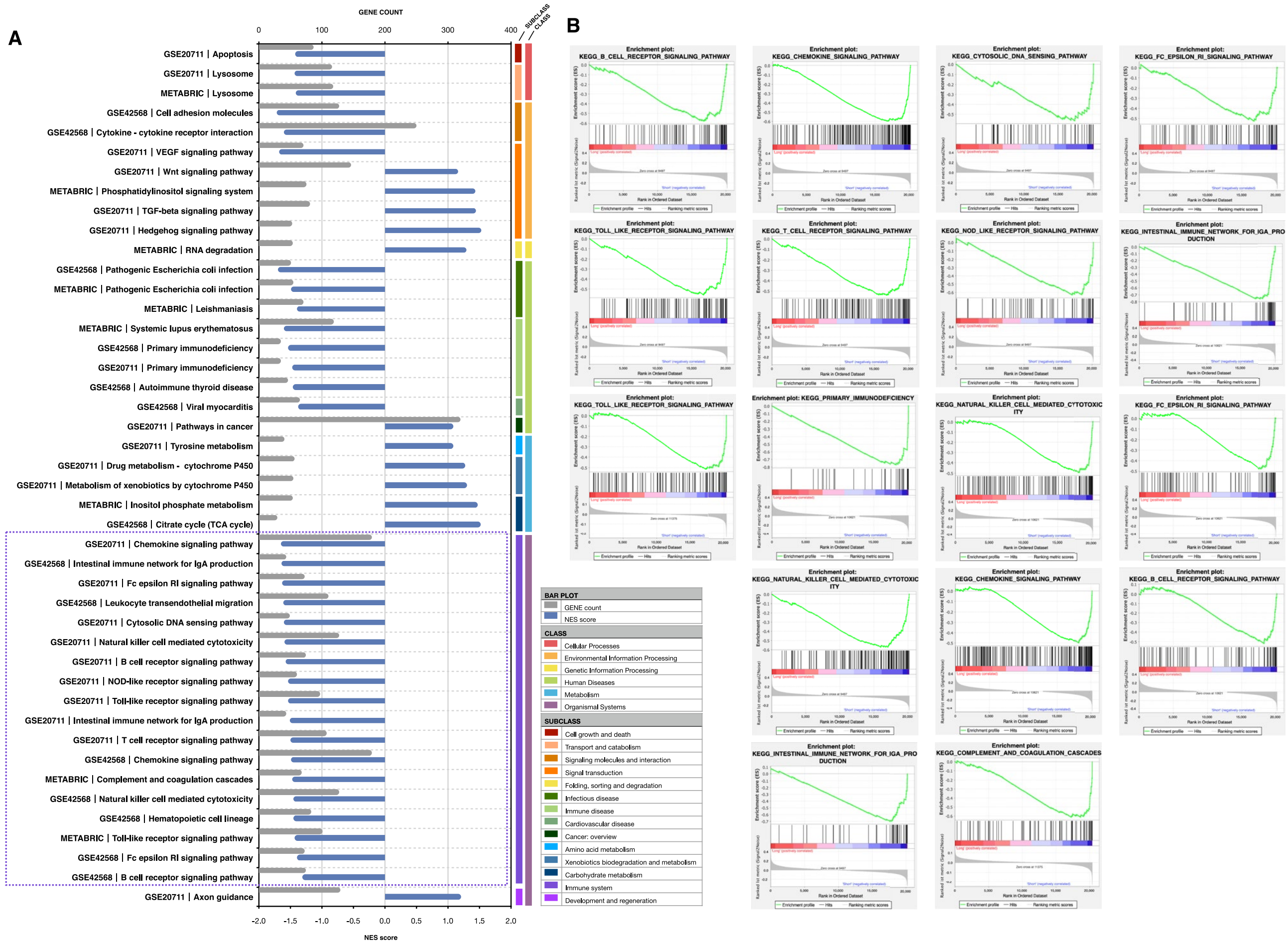


Fig. 2. Pathways associated with tumor recurrence via GSEA analysis. (**A**) The bar chart illustrates the primary enrichment pathways of the short-term (ST) and long-term (LT) groups. Blue bars represent normalized enrichment scores (NES), and gray lines indicate the number of genes enriched in each pathway. The positive and negative X-axis coordinates correspond to the enrichment pathways of the LT group and ST group, respectively. The color bars on the right side of the chart represent the classification groups of pathways. **(B)** The top pathways in the ST group were found to be significantly enriched, highlighting their potential relevance to tumor recurrence mechanisms.

(METABRIC and TCGA) but poorly on validation datasets, IRGS maintains its performance across different datasets, suggesting better generalization and less overfitting. This is likely due to our rigorous machine learning algorithms and dimensionality reduction techniques, which enhance its potential for external inference.

Relationship between IRGS and patient outcomes and clinical factors

Our evaluation of IRGS's relationship with patient survival outcomes and clinical indices revealed several key insights. In terms of outcome (OS/RFS/DSS/DFI), the risk score was significantly lower in in patients who survived without recurrence or progression compared to those who died, recurred, or progressed (Fig. 5A). Patients at advanced stages of disease progression exhibited significantly higher RS than those at earlier stages (Fig. 5B). Significant variations in RS were observed among different breast cancer subtypes (Fig. 5C). Pairwise comparisons in the TCGA and METABRIC datasets showed that Normal-like subtype had the lowest RS, with significant differences from other subtypes. This suggests that IRGS can effectively distinguish risk profiles across different subgroups, providing valuable insights for personalized risk assessment and clinical decision-making.

Biological functions and pathways influenced by IRGS

To explore the biological functions and pathways in which IRGS participate and its influence on tumor progression, we performed differential gene expression analysis and annotation on high- and low-risk groups from the TCGA dataset. The results revealed 300 up-regulated and 564 down-regulated in the high-risk group compared to the low-risk group (Fig. 6A). These differentially expressed genes were mainly enriched in KEGG pathways such as cytokine-cytokine receptor interaction (hsa04060, $p=$ 2.69E-13), cell cycle (hsa04110, $p=$ 2.49E-10), chemokine signaling pathway (hsa04062, $p=$ 8.38E-06), PI3 K-Akt signaling pathway (hsa04151, $p=$ 0.003), and PD-L1 expression and PD-1 checkpoint pathway in cancer (hsa05235, $p=$ 0.004) (Fig. 6B, Table S5). The GO analysis showed significant enrichment of immune-related annotations in the low-risk group, including immune response-regulating cell surface receptor signaling pathway (GO:0,002,768, $p = 2.09E-24$), immunological synapse (GO:0,001,772, $p = 6.91E-05$), and immune receptor activity (GO:0,140,375, $p = 2.65E-17$). In contrast, the high-risk group exhibited significant enrichment of cell cycle-related GO annotations, such as nuclear chromosome segregation (GO:0,098,813, $p = 3.49E-19$), chromosomal region (GO:0,098,687, $p = 2.49E-12$), and microtubule binding (GO:0,008,017, $p = 2.16E-05$) (Fig. 6C, Table S6). These findings indicate that IRGS may play a crucial role in regulating tumor progression through these biological functions and pathways.

Evaluation of IRGS on the immune landscape

Next, we used the ESTIMATE algorithm to assess the TME in high- and low-risk patient groups in the TCGA cohort, based on the IRGS risk assessment system, and systematically compared immune function differences between the two groups. Results showed that the stromal score, immune score, ESTIMATE score were markedly higher in the low-risk group than in the high-risk group, while tumor purity was significantly lower ($p < 0.001$, Fig. 6D). This suggests that high-risk samples have higher tumor purity and lower immune cell infiltration.

To comprehensively evaluate tumor immune cell infiltration differences, we used five algorithms (TIMER, CIBERSORT, QUANTISEQ, MCPcounter, and EPIC) to quantify immune cell infiltration in patient samples based on the gene expression profiles of 22 immune cell markers and analyzed the correlation between IRGS and immune cell infiltration. In the low-risk group, the proportion of lymphocytes, myeloid cells, and innate immune cells, such as B cells, CD8 + T cells, CD4 + memory T cells, NK cells, monocytes, and M1/M2 macrophages, was significantly higher than in the high-risk group ($p < 0.05$; Fig. 6E). Figure 6F presents a heatmap showing the correlation between IRGS gene expression and immune cell infiltration ($p < 0.005$). Based on correlation patterns, IRGS genes were categorized into three classes: Class I includes MS4A1, HZMB, FOS, and CDC42, which show a strong positive correlation with immune infiltrating cells; Class II includes TANK, BLINK, CD1C, and CHUK, which show a weak positive correlation with immune infiltrating cells; Class III includes TXN, MAP2K4, SERPINA5, and VAV3, which show a negative correlation with immune infiltrating cells.

IRGS as a predictive biomarker for immunotherapy in breast cancer patients

Given the link between IRGS and immune cell infiltration, we compared the expression levels of common immune suppressor between high- and low-risk groups (Fig. 7A). Results showed that PDL1, PD1, TIGIT, LAG3, and CTLA4 were up-regulated in the low-risk group ($p < 0.05$). Using the TIDE algorithm, we predicted immune responses in the METABRIC and TCGA cohorts to see if IRGS could predict immunotherapy response. There was a significant difference in TIDE scores between the groups, with high-risk patients having higher scores ($p < 0.01$; Fig. 7B). Patients were divided into responder and non-responder groups based on TIDE scores. PDL1 (CD274) expression was higher in responders, while CAF, dysfunction, and exclusion scores were lower ($p < 0.01$; Fig. 7C). This suggests immune checkpoint inhibitors may be less effective in high-risk patients. Thus, IRGS can serve as a predictive biomarker for breast cancer immunotherapy.

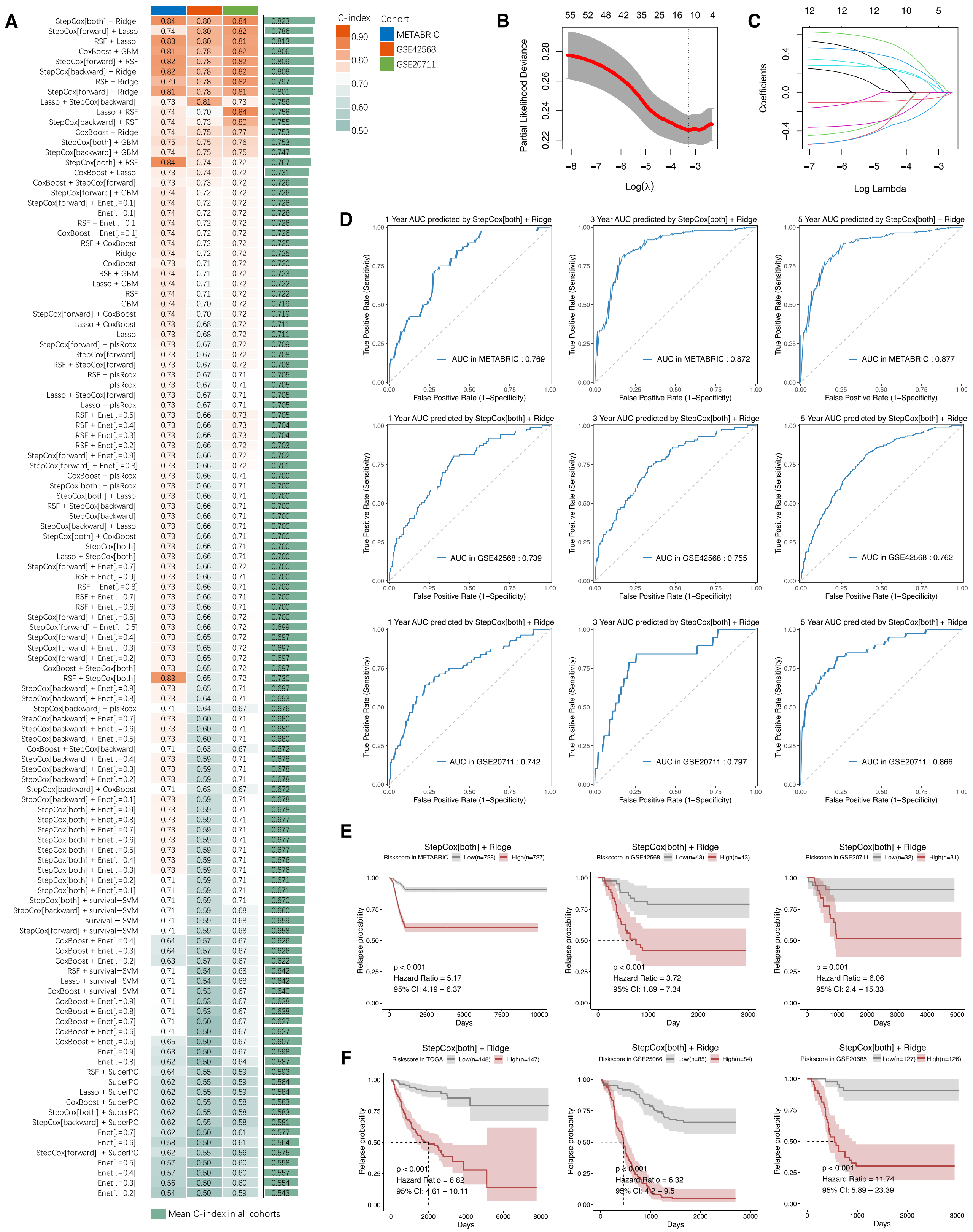
Chemotherapeutic sensitivity prediction

We analyzed IC50 values to assess drug responsiveness in high- and low-risk groups and identify suitable drugs for each group. As shown in Fig. 7D ($p < 0.05$), the high-risk/non-responder group had had lower IC50 values to Crizotinib and Lisinopril, indicating greater sensitivity to these drugs. Conversely, the low-risk/responder group had lower IC50 values responding to seven drugs, including Fluorouracil, Bleomycin, and Gemcitabine, suggesting they are more effective in this group.

Discussion

In the era of precision medicine, relying solely on traditional indicators such as AJCC TNM staging and biochemical markers is insufficient for accurately assessing patient prognosis. Similarly, evaluating drug resistance or relapse risk based on these methods remains insufficiently effective. Existing tumor models often depend on single-feature selection and simplistic algorithms, lacking rigorous validation across multiple datasets. This limitation frequently results in suboptimal performance of prognostic biomarkers or overfitting to specific datasets. To address these challenges, this study leveraged 10 widely used machine learning algorithms to construct 117 models based on multi-dataset integration from RNA-seq high-throughput sequencing technology. Through systematic analysis, we identified a 12-gene immune-related gene signature (IRGS) for breast cancer recurrence prediction. After thorough comparison and validation, the combination of StepCox[both] and Ridge regression emerged as the optimal model for determining the IRGS.

StepCox regression constructs an optimal prediction model by iteratively selecting variables. The StepCox[both] mode integrates backward and forward strategies, allowing the model to dynamically add or remove variables in each iteration until reaching a stable state where no further additions or removals are possible. This approach automatically identifies variables significantly associated with survival time. Ridge regression then refines the model by applying L2 regularization to optimize variable weights, mitigating overfitting inherent in stepwise Cox regression. Together, these algorithms enhance model stability and predictive accuracy. For instance, in head and neck cancer research, a prognostic model with 44 genes was built using StepCox[both] and ridge regression to predict patient prognosis and immune features[46]. In bladder cancer studies, a cell death gene-based signature was developed via StepCox[backward] and ridge regression[47]. In ovarian cancer, a prognostic signature comprising 20 exosome-derived lncRNAs was constructed using StepCox[forward] and ridge regression algorithms[48]. The IRGS signature in this study demonstrated exceptional predictive performance across multiple independent datasets and outperformed 10 previously published biomarkers in terms of prediction accuracy. Given its robustness and generalizability, the IRGS identified in this study can serve as a valuable auxiliary tool for clinicians to assess breast cancer recurrence risk, potentially enhancing precision in patient management and therapeutic decision-making.



Based on the correlation between tumor immune cell infiltration and IRGS expression, the 12-gene signature can be categorized into three distinct classes. Class I includes genes such as MS4A1, HZMB, FOS, and CDC42, which are strongly positively correlated with immune infiltration. Class II consists of genes like TANK, BLNK, CD1 C, and CHUK, which show weak positive correlations with immune infiltration. Class III includes genes such as TXN, MAP2K4, SQSTM1, and VAV3, which are negatively correlated with immune infiltration. Patients classified as low-risk based on the IRGS exhibited significantly higher immune scores compared to high-risk

◀**Fig. 3**. Development and validation of IRGS via an integrated machine learning program. (**A**) A total of 117 prediction models were constructed using the LOOCV framework, with the C-index calculated for each model across all validation datasets. (**B**) In the METABRIC cohort, the data are presented as the mean ± 95%CI for the C-coefficient derived from stepwise Cox regression. (**C**) The optimum λ is determined at the point of minimal partial likelihood deviation, and the Ridge regression coefficients for the most informative prognostic genes were subsequently generated. (**D**) The 1, 3, and 5-year time-dependent ROC curve were evaluated in the METABRIC, GSE20711, and GSE42568 cohorts. (**E** and **F**) Kaplan–Meier curve illustrating RFS of patients according to the IRGS signature in training, validation, and external cohorts.

patients. This correlation was strongly associated with elevated infiltration of key immune cell types, including CD8 + T cells, CD4 + memory T cells, follicular helper T cells (Tfh), and M1/M2 macrophages. CD4 + memory T cells play a central role in adaptive immunity by generating robust effector responses, promoting B cell proliferation, rapidly activating antigen-presenting cells, and producing cytokines such as IL-2 to enhance CD8 + T cell activity[49–51]. Tfh, a specialized subset of CD4 + T cells, primarily function in germinal centers to support B cells during humoral immune responses. In the TME, Tfh cells are linked to improved cancer prognosis, and in breast cancer, their presence correlates with effective adaptive immune responses and better patient survival[52]. M1 and M2 macrophages have opposing roles in tumor progression: M1 macrophages suppress tumor growth and enhance anti-tumor immunity, while M2 macrophages promote tumor growth, metastasis, and immune evasion by inhibiting immune responses. In this study, low-risk patients exhibited high infiltration of CD4 +, CD8 +, Tfh, and M1 macrophages, indicating a strong immune response capable of suppressing tumor progression. However, these patients also showed elevated infiltration of M2 macrophages. This dual pattern suggests that while the low-risk group currently benefits from a robust anti-tumor immune response, the presence of M2 macrophages may pose a latent risk of immune evasion, potentially undermining long-term therapeutic outcomes. This comprehensive analysis highlights the complex interplay between immune cell infiltration and tumor progression, underscoring the importance of integrating IRGS with immune microenvironment dynamics for personalized risk assessment and therapeutic strategies in breast cancer.

The high expression of immune checkpoint molecules in the tumor microenvironment plays a pivotal role in suppressing anti-tumor immunity, particularly through mechanisms involving PD-L1. Studies have demonstrated that PD-L1 expression on the surface of breast cancer cells is closely associated with patient prognosis. Notably, PD-L1-positive breast cancer patients, especially those with triple-negative breast cancer (TNBC), exhibit higher pathological complete response (pCR) rates and relatively better outcomes when treated with immune checkpoint inhibitors[53]. Furthermore, research has shown that immune checkpoint-related gene markers can serve as predictors of a patient's response to immunotherapy[54]. In this context, patients in the high-risk group may exhibit poorer responses to immune checkpoint inhibitors, whereas those in the low-risk group are more likely to benefit from such therapies. In this study, we observed significantly lower expression levels of immune checkpoint genes in the high-risk group compared to the low-risk group. Combined with TIDE analysis, these findings suggest that patients in the low-risk group may be more responsive to immune checkpoint inhibitors. This highlights the potential clinical utility of integrating immune checkpoint expression profiles with prognostic biomarkers like the IRGS to guide personalized immunotherapy strategies in breast cancer.

Through sensitivity analysis of chemotherapy and targeted drugs, we can explore more effective potential drugs for high- and low-risk patient groups. In cancers with a high incidence of p53 mutations, such as TNBC and high-grade serous ovarian cancer (HGSOC), crizotinib shows potential in combination therapy[55]. The combination of focal adhesion kinase (FAK) inhibitor IN10018 and crizotinib exhibits synergistic anti-tumor effects by upregulating the p53 signaling pathway and inhibiting FAK and ROS1[56]. Trastuzumab (TZB) is an established treatment for HER2-positive breast cancer. However, TZB often causes cardiac toxicity, which can affect heart function. Lisinopril can prevent cardiac toxicity caused by TZB in patients with locally advanced breast cancer, allowing them to undergo complete and uninterrupted TZB treatment[57]. A hyaluronic acid-based drug combination of imiquimod and gemcitabine shows enhanced anti-cancer activity against breast tumor cells in vitro and significantly inhibits the volume of 4T1 breast tumors in mice, with higher CD68 + tumor-associated macrophage infiltration in the tumors[58]. Another study found that compared to free drugs, liposomal co-delivery of rapamycin and resveratrol showed higher cytotoxicity and significant anti-tumor potential against breast cancer cells[59]. The STAT3 inhibitor pyrimethamine demonstrates anti-cancer and immune-stimulating effects in a breast cancer mouse model. Treatment of breast cancer tumor-transplanted mice with pyrimethamine showed reduced STAT3 activation in tumor cells, weakened tumor growth, and decreased tumor-associated inflammation, suggesting that pyrimethamine has direct and indirect anti-cancer and immune-stimulating effects[60]. Single-agent chemotherapy regimens face significant barriers in clinical efficacy. The complexity of cancer highlights the need for an alternative approach that combines two or more therapeutic strategies to win the battle. The potential effective drugs and corresponding immunotherapies for patients with different risk profiles explored in this study will provide treatment options for the development of efficient, low-toxicity, and personalized drug combinations.

The IRGS model offers a promising approach for clinical translation and implementation due to its simplicity and reliance on PCR testing. However, several limitations should be acknowledged. First, the samples used in this study were retrospective, and the IRGS should be validated in future prospective multicenter cohorts to ensure its reliability and generalizability. Second, the clinical and molecular features available in public datasets are insufficient, which may obscure potential associations between the IRGS and specific variables. Third, the roles of most of the genes in the IRGS in breast cancer are not yet fully understood and require further in vivo

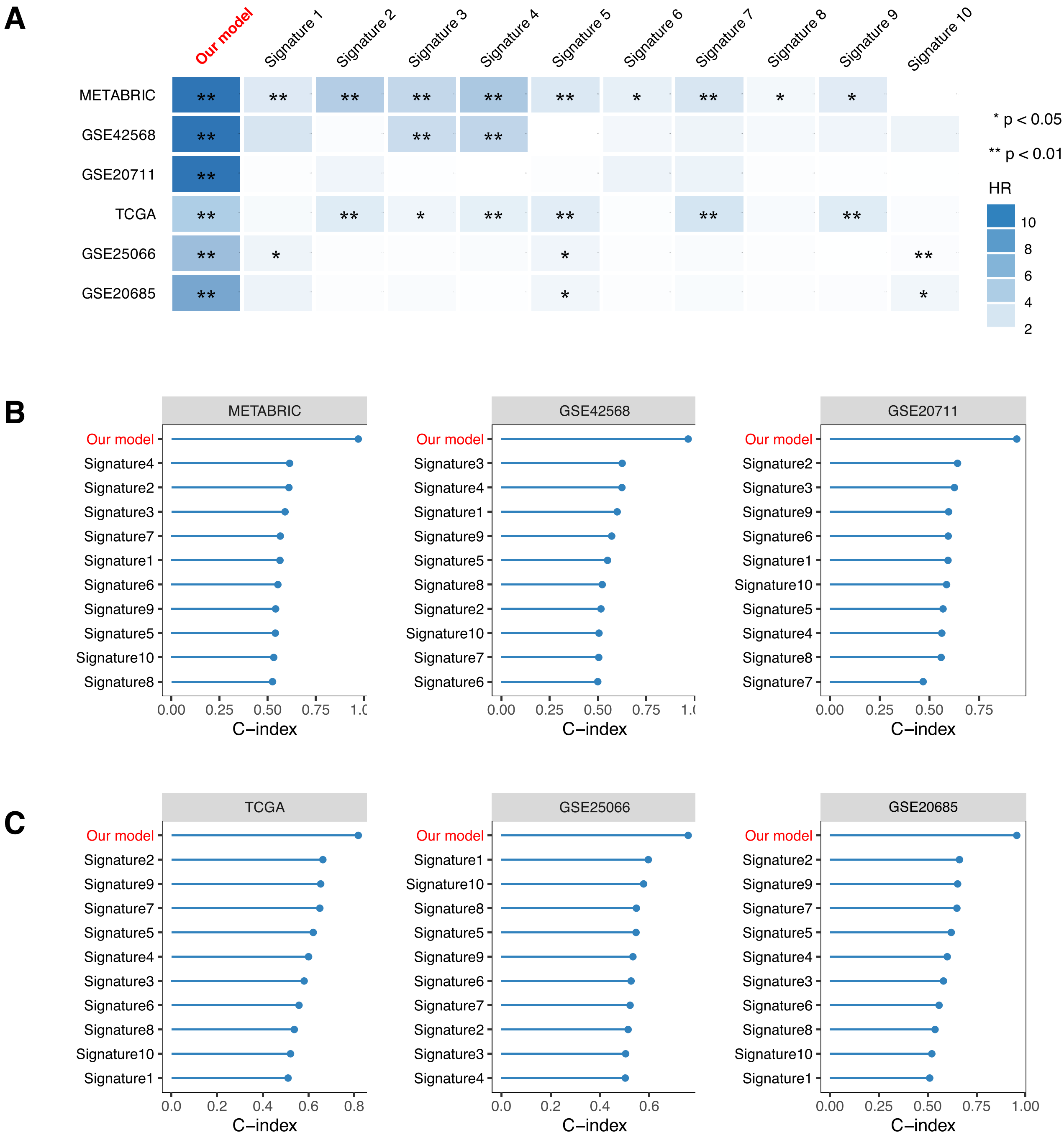


Fig. 4. Comparison of prognostic signatures based on gene expression in breast cancer. (**A**) Univariate Cox regression analysis was performed for IRGS and ten previously published signatures (PMID:34,059,009, PMID:35,443,644, PMID:37,521,466, PMID:37,685,980, PMID:36,993,976, PMID:36,226,193, PMID:38,074,669, PMID:33,584,825, PMID:33,948,357 and PMID:35,646,948). HR and p-values were compared to evaluate the prognostic performance of each signature. (**B**) The C index was calculated and compared for the IRGS and the ten published signatures, which collectively encompass 82 genes. Statistical comparisons were performed using a bilateral Z-score test. Data are presented as mean ± 95%CI. $^{*}p < 0.05$; $^{**}p < 0.01$.

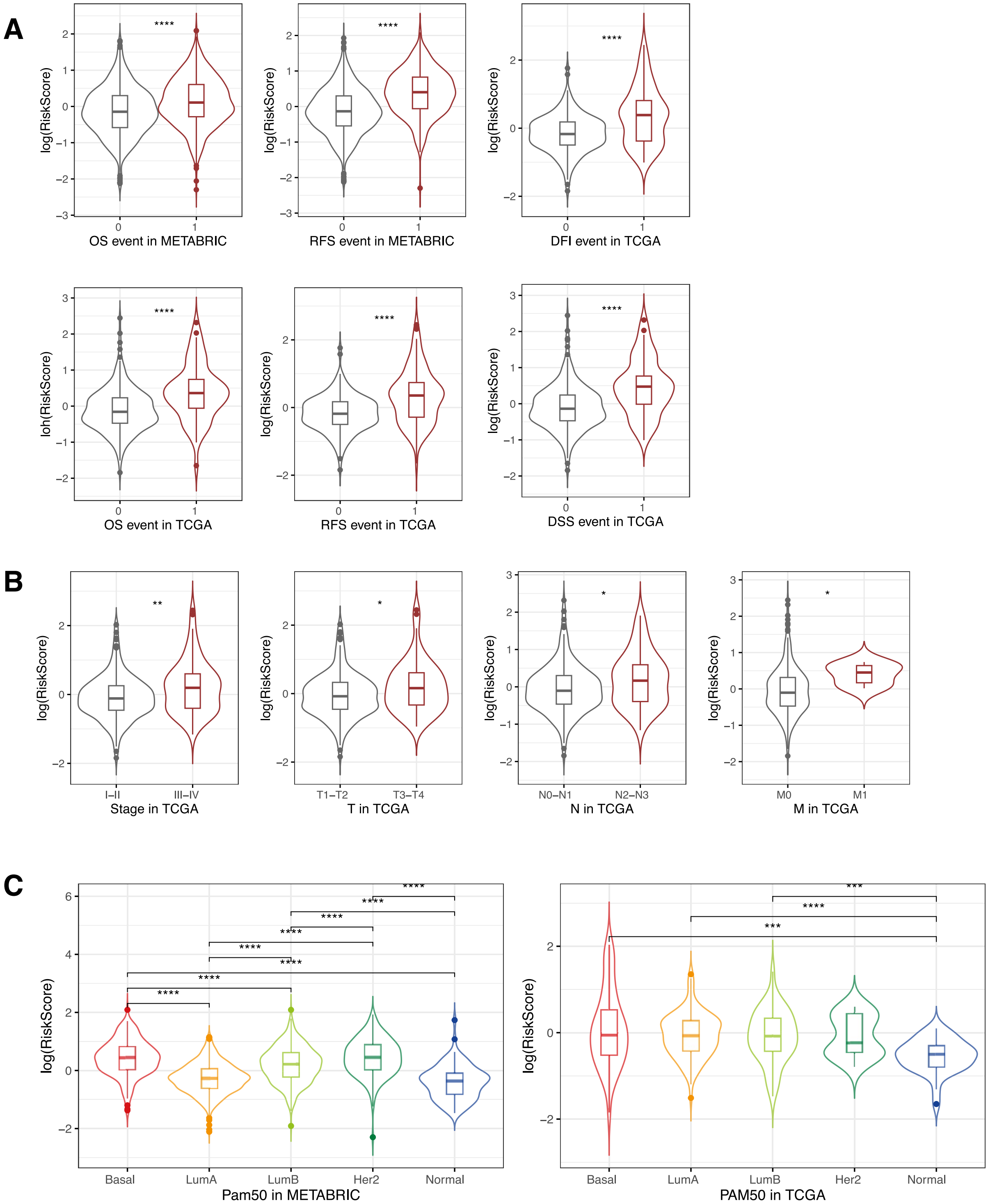


Fig. 5. The relationship between IRGS and the clinical characteristics of patients. Kaplan–Meier (KM) analysis was performed to evaluated the relationship between high- and low-risk groups and clinical subgroups within the METABRIC and TCGA cohorts. The differences in risk score across the clinicopathologic subgroups were analyzed. Additionally, the expression levels of prognostic signature genes were compared between high- and low-risk groups cross different outcomes (**A**), clinical subgroups (**B**), and Pam50 subtypes (**C**). Statistical analysis was conducted using Analysis of Variance (ANOVA) followed by Tukey's honestly significant difference (HSD) test for groups with more than two subgroups and the Wilcoxon test for groups with two subgroups. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.005$.

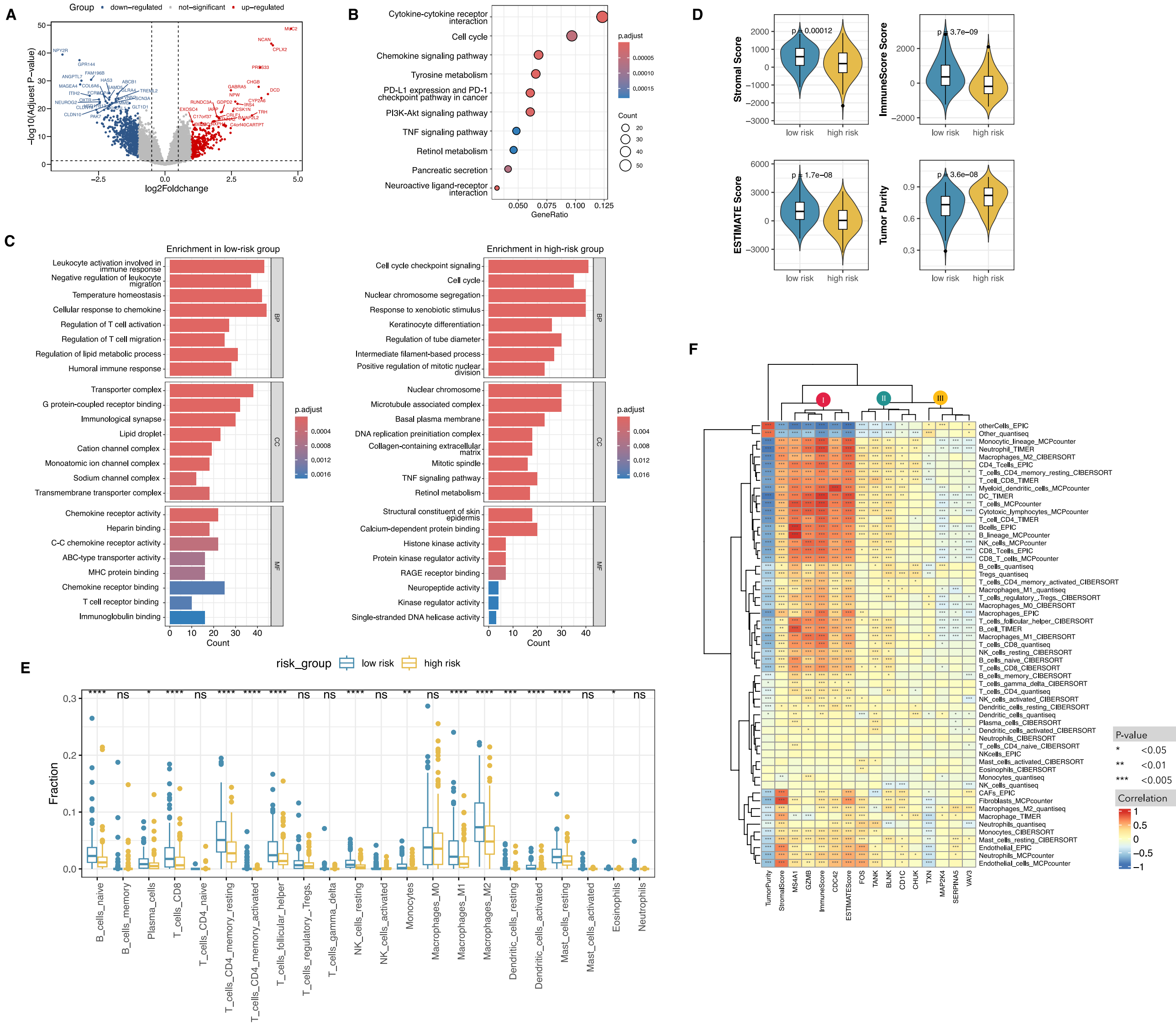


Fig. 6. IRGS associated with pathways and immune infiltration. (**A**) A volcano plot illustrates differentially expressed genes (DEGs) between high- and low-risk groups, with red and blue dots indicating the up-regulated and down-regulated DEGs, while as the grey dots represent non-significant genes. (**B**) KEGG pathway enrichment analysis of DEGs highlights significantly enriched signaling pathways. (**C**) GO enrichment analysis of DEGs reveals key biological processes, cellular components, and molecular functions. (**D**) ESTIMATE analysis was used to assess and compare immune score, stromal score, tumor purity, and the composite ESTIMATE score between high- and low-risk groups. (**E**) Differences in the abundance of tumor-infiltrating immune cells between high- and low-risk groups were analyzed using the CIBERSORT algorithm. (**F**) Correlation matrices generated using TIMER, CIBERSORT, quantiseq, MCPcounter, and EPIC tools illustrate relationships between risk scores, IRGS genes, immune scores, stromal scores, ESTIMATE scores, and abundance of 22 immune-infiltrating cells. Red indicates positive correlation, while blue indicates negative correlation. *$p < 0.05$, **$p < 0.01$, ***$p < 0.005$.

and in vitro experiments to elucidate their functions and mechanisms. Addressing these limitations will be critical for optimizing the IRGS and enhancing its clinical utility.

Conclusion

This study conducted a systematic analysis of RNA-seq high-throughput data, utilizing a combination of 10 machine learning algorithms to construct 117 models. The StepCox[both] and ridge regression were jointly assessed as the optimal algorithm, ultimately identifying an immune-related gene signature (IRGS) consisting of 12 genes. The IRGS demonstrated outstanding predictive performance across multiple datasets and surpassed 10 previously published signatures. Our signature exhibits potential in predicting responses to immunotherapy and chemotherapy. Upon clinical validation, the IRGS has the capacity to enhance personalized treatment planning by identifying patients at risk of relapse for informed decision-making. In conclusion, the IRGS can serve as an

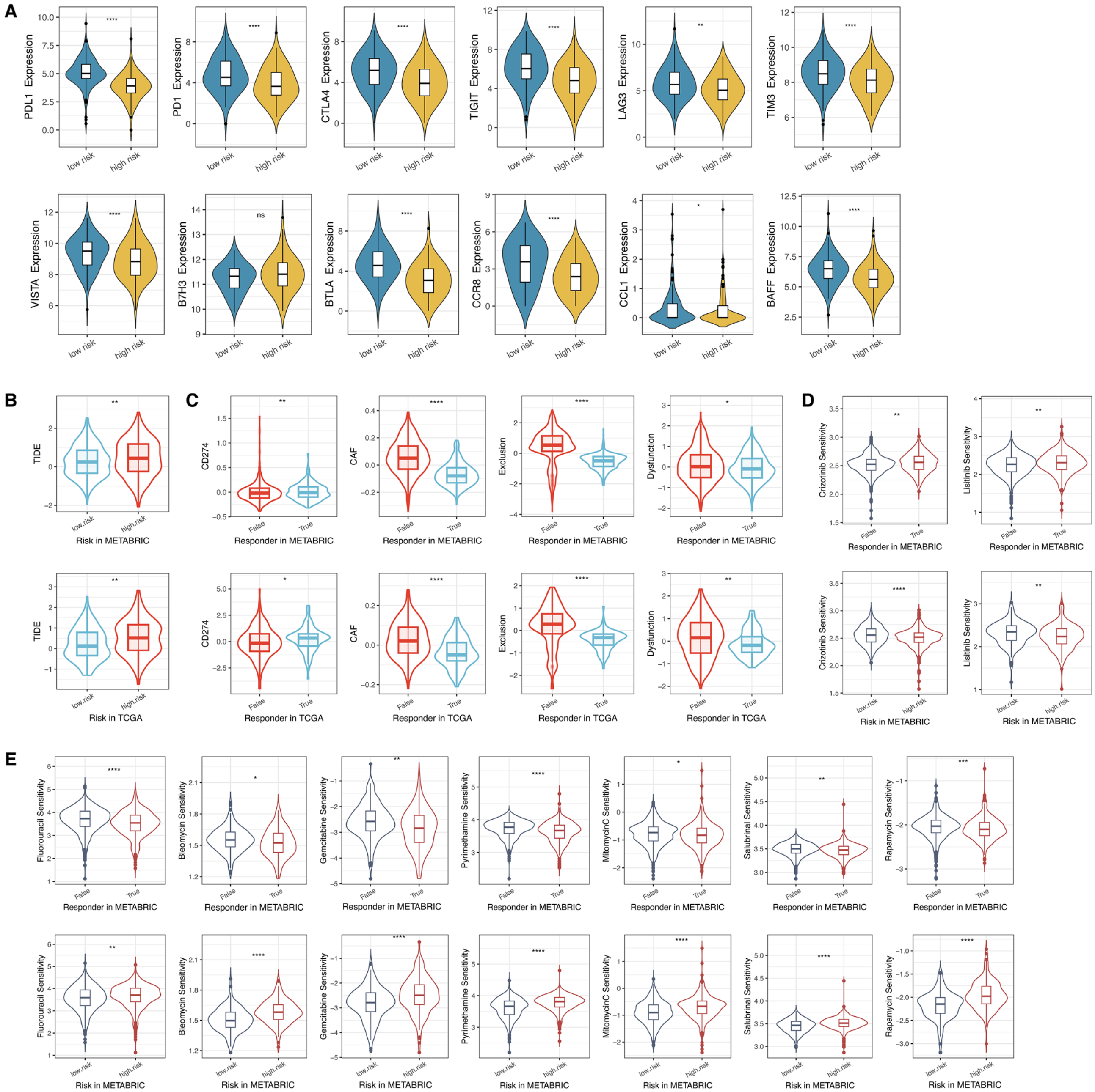


Fig. 7. Evaluation of immunotherapy response and drug sensitivity by IRGS. (**A**) A comparison of the expression levels of immune checkpoint inhibitor (ICI) genes between high- and low-risk groups. (**B**) TIDE scores were calculated to evaluate the response of high- and low-risk groups to ICI therapy. (**C**) Immune differences between response and non-response groups were analyzed based on TIDE evaluation. (**D**) Sensitive drugs for non-responders and high-risk patients in immunotherapy were identified. (**E**) Sensitive drugs for responders and low-risk patients in immunotherapy were identified. ns, not significant, $*p < 0.05$, $**p < 0.01$, $***p < 0.005$.

adjunct for assessing the risk of breast cancer recurrence, thereby improving the clinical management of patients with breast cancer.

Data availability

The datasets generated during the current study are available in the TCGA (https://xenabrowser.net), METAB RIC (http://www.cbioportal.org/) and GEO (https://www.ncbi.nlm.nih.gov/geo/) public databases. The patients involved in the database have obtained ethical approval. Users can download relevant data for free for research and publish relevant articles. Our study is based on open-source data, so there are no ethical issues and other conflicts of interest.

Received: 18 September 2024; Accepted: 20 May 2025
Published online: 05 June 2025

References

1. Bray, F. et al. Global cancer statistics 2018: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA Cancer J Clin.* **68**, 394–424 (2018).
2. Maughan, K. L., Lutterbie, M. A. & Ham, P. S. Treatment of breast cancer. *Am. Fam. Physician.* **81**, 1339–1346 (2010).
3. Siegel, R. L., Miller, K. D., Fuchs, H. E. & Jemal, A. Cancer statistics, 2021. *CA A Cancer J. Clinicians* **72**(1), 7–33 (2021).
4. Shao, F. & Fitzgerald, K. A. Molecular mechanisms and functions of pyroptosis. *J. Mol. Biol.* **434**, 167461 (2022).
5. Chen, X. & Cubillos-Ruiz, J. R. Endoplasmic reticulum stress signals in the tumour and its microenvironment. *Nat. Rev. Cancer.* **21**, 71–88 (2021).
6. Cong, K. et al. Replication gaps are a key determinant of PARP inhibitor synthetic lethality with BRCA deficiency. *Mol. Cell.* **81**, 3128-3144.e7 (2021).
7. Morganti, S. et al. Adjuvant olaparib for germline BRCA carriers with HER2-negative early breast cancer: Evidence and controversies. *Oncologist.* **28**, 565–574 (2023).
8. Robson, M. E. et al. OlympiAD final overall survival and tolerability results: Olaparib versus chemotherapy treatment of physician's choice in patients with a germline BRCA mutation and HER2-negative metastatic breast cancer. *Ann. Oncol.* **30**, 558–566 (2019).
9. Hawley, S. T. et al. Recurrence risk perception and quality of life following treatment of breast cancer. *Breast Cancer Res Treat.* **161**, 557–565 (2017).
10. Garcia-Murillas, I. et al. Assessment of molecular relapse detection in early-stage breast cancer. *JAMA Oncol.* **5**, 1473–1478 (2019).
11. Biggar, R. J., Andersen, E. W., Kroman, N., Wohlfahrt, J. & Melbye, M. Breast cancer in women using digoxin: tumor characteristics and relapse risk. *Breast Cancer Res.* **15**, R13 (2013).
12. Wang, F. et al. Association of serum cortisol and cortisone levels and risk of recurrence after endocrine treatment in breast cancer. *Clin. Exp. Med.* **23**, 3883–3893 (2023).
13. Wang, S.-Y., Shamliyan, T., Virnig, B. A. & Kane, R. Tumor characteristics as predictors of local recurrence after treatment of ductal carcinoma in situ: A meta-analysis. *Breast Cancer Res. Treat.* **127**, 1–14 (2011).
14. Zhang, Y. et al. Breast cancer index identifies early-stage estrogen receptor-positive breast cancer patients at risk for early- and late-distant recurrence. *Clin. Cancer Res.* **19**, 4196–4205 (2013).
15. Lee, Y. J. et al. Prognosis according to the timing of recurrence in breast cancer. *Ann. Surg. Treat. Res.* **104**, 1–9 (2023).
16. Chua, A. V. et al. Epidemiology of early vs late recurrence among women with early stage estrogen receptor-positive breast cancer in the pathways study. *J. Natl. Cancer Inst.* **116**, 1621–1631 (2024).
17. Ritchie, M. E. et al. limma powers differential expression analyses for RNA-sequencing and microarray studies. *Nucleic Acids Res.* **43**, e47 (2015).
18. Kanehisa, M. & Goto, S. KEGG: Kyoto encyclopedia of genes and genomes. *Nucleic Acids Res.* **28**, 27–30 (2000).
19. Zou, H. & Hastie, T. Regularization and variable selection via the elastic net. *J. R. Stat. Soc. Ser. B Stat. Methodol.* **67**, 301–320 (2005).
20. Arashi, M., Roozbeh, M., Hamzah, N. A. & Gasparini, M. Ridge regression and its applications in genetic studies. *PLoS ONE* **16**, e0245376 (2021).
21. Bair, E. & Tibshirani, R. Semi-supervised methods to predict patient survival from gene expression data. *PLoS Biol.* **2**, e108 (2004).
22. Wang, H. & Zhou, L. Random survival forest with space extensions for censored data. *Artif. Intell. Med.* **79**, 52–61 (2017).
23. Lee, D., Lee, Y., Pawitan, Y. & Lee, W. Sparse partial least-squares regression for high-throughput survival data analysis. *Stat. Med.* **32**, 5340–5352 (2013).
24. Emblem, K. E. et al. A generic support vector machine model for preoperative glioma survival associations. *Radiology* **275**, 228–234 (2015).
25. Zhang, L., Zhou, W. & Jiao, L. Wavelet support vector machine. *IEEE Trans. Syst. Man Cybern. B Cybern.* **34**, 34–39 (2004).
26. Binson, V. A. et al. A review of machine learning algorithms for biomedical applications. *Ann. Biomed. Eng.* **52**, 1159–1183 (2024).
27. Li, Y., Wu, X., Yang, P., Jiang, G. & Luo, Y. Machine learning for lung cancer diagnosis, treatment, and prognosis. *Genom. Proteom. Bioinform.* **20**, 850–866 (2022).
28. Wang, D. et al. Identification of the prognostic value of ferroptosis-related gene signature in breast cancer patients. *BMC Cancer* **21**, 645 (2021).
29. Chen, H., Luo, H., Wang, J., Li, J. & Jiang, Y. Identification of a pyroptosis-related prognostic signature in breast cancer. *BMC Cancer* **22**, 429 (2022).
30. Li, J. et al. Cuproptosis/ferroptosis-related gene signature is correlated with immune infiltration and predict the prognosis for patients with breast cancer. *Front. Pharmacol.* **14**, 1192434 (2023).
31. Huang, B. et al. Construction of a matrix cancer-associated fibroblast signature gene-based risk prognostic signature for directing immunotherapy in patients with breast cancer using single-cell analysis and machine learning. *Int. J. Mol. Sci.* **24**, 13175 (2023).
32. Luo, Y. et al. A degradome-based prognostic signature that correlates with immune infiltration and tumor mutation burden in breast cancer. *Front. Immunol.* **14**, 1140993 (2023).
33. Xing, Z. et al. Construction of a prognostic 6-gene signature for breast cancer based on multi-omics and single-cell data. *Front. Oncol.* **13**, 1186858 (2023).
34. Zhang, X. et al. A novel glycolysis-related four-mRNA signature for predicting the survival of patients with breast cancer. *Front. Genet.* **12**, 606937 (2021).
35. Zhang, Y. et al. An immune-related signature that to improve prognosis prediction of breast cancer. *Am. J. Cancer Res.* **11**, 1267–1285 (2021).
36. Chu, L. et al. A prognostic signature consisting of pyroptosis-related genes and SCAF11 for predicting immune response in breast cancer. *Front. Med. (Lausanne).* **9**, 882763 (2022).
37. Yu, G., Wang, L.-G., Han, Y. & He, Q.-Y. clusterProfiler: An R package for comparing biological themes among gene clusters. *OMICS* **16**, 284–287 (2012).
38. Yoshihara, K. et al. Inferring tumour purity and stromal and immune cell admixture from expression data. *Nat. Commun.* **4**, 2612 (2013).
39. Zhang, Y. et al. Multi-omics data analyses construct TME and identify the immune-related prognosis signatures in human LUAD. *Mol. Ther. Nucleic Acids.* **21**, 860–873 (2020).
40. Li, T. et al. TIMER: A web server for comprehensive analysis of tumor-infiltrating immune cells. *Cancer Res.* **77**, e108–e110 (2017).
41. Newman, A. M. et al. Robust enumeration of cell subsets from tissue expression profiles. *Nat. Methods.* **12**, 453–457 (2015).
42. Finotello, F. et al. Molecular and pharmacological modulators of the tumor immune contexture revealed by deconvolution of RNA-seq data. *Genome Med.* **11**, 34 (2019).
43. Becht, E. et al. Estimating the population abundance of tissue-infiltrating immune and stromal cell populations using gene expression. *Genome Biol.* **17**, 218 (2016).
44. Jiang, P. et al. Signatures of T cell dysfunction and exclusion predict cancer immunotherapy response. *Nat. Med.* **24**, 1550–1558 (2018).

45. Yang, W. et al. Genomics of drug sensitivity in cancer (GDSC): A resource for therapeutic biomarker discovery in cancer cells. *Nucleic Acids Res.* **41**, D955-961 (2013).
46. Fang, X.-L. et al. Identification of a gene score related to antigen processing and presentation machinery for predicting prognosis in head and neck squamous cell carcinoma and its potential implications for immunotherapy. *Clin. Transl. Oncol.* https://doi.org/10.1007/s12094-024-03829-2s (2024).
47. Wang, Y. & Zhang, Q. Leveraging programmed cell death signature to predict clinical outcome and immunotherapy benefits in postoperative bladder cancer. *Sci. Rep.* **14**, 22976 (2024).
48. Cui, Y. et al. An exosome-derived lncRNA signature identified by machine learning associated with prognosis and biomarkers for immunotherapy in ovarian cancer. *Front. Immunol.* **15**, 1228235 (2024).
49. Krueger, P. D., Osum, K. C. & Jenkins, M. K. CD4+ memory t-cell formation during type 1 immune responses. *Cold Spring Harb. Perspect. Biol.* **13**, a038141 (2021).
50. Künzli, M. & Masopust, D. CD4+ T cell memory. *Nat. Immunol.* **24**, 903–914 (2023).
51. Raphael, I., Joern, R. R. & Forsthuber, T. G. Memory CD4+ T cells in immunity and autoimmune diseases. *Cells* **9**, 531 (2020).
52. Sun, L., Su, Y., Jiao, A., Wang, X. & Zhang, B. T cells in health and disease. *Signal Transduct. Target Ther.* **8**, 235 (2023).
53. Zdenkowski, N. et al. Timing of nivolumab with neoadjuvant carboplatin and paclitaxel for early triple-negative breast cancer (BCT1902/IBCSG 61–20; Neo-N): a non-comparative, open-label, randomised, phase 2 trial. *Lancet Oncol.* **26**, 367–377 (2025).
54. Yin, K. et al. Identification of a novel immune checkpoint-related gene signature predicts prognosis and immunotherapy in breast cancer and experiment verification. *Sci. Rep.* **14**, 31065 (2024).
55. Musa, S. et al. Overcoming chemoresistance in Cancer: The promise of crizotinib. *Cancers (Basel).* **16**, 2479 (2024).
56. Tan, X. et al. Simultaneous inhibition of FAK and ROS1 synergistically repressed triple-negative breast cancer by upregulating p53 signalling. *Biomark. Res.* **12**, 13 (2024).
57. Guglin, M., Munster, P., Fink, A. & Krischer, J. Lisinopril or Coreg CR in reducing cardiotoxicity in women with breast cancer receiving trastuzumab: A rationale and design of a randomized clinical trial. *Am. Heart J.* **188**, 87–92 (2017).
58. Singh, B. et al. Imiquimod-gemcitabine nanoparticles harness immune cells to suppress breast cancer. *Biomaterials* **280**, 121302 (2022).
59. Dos Reis, L. R. et al. Design of rapamycin and resveratrol coloaded liposomal formulation for breast cancer therapy. *Nanomedicine (Lond).* **18**, 789–801 (2023).
60. Khan, M. W. et al. The STAT3 inhibitor pyrimethamine displays anti-cancer and immune stimulatory effects in murine models of breast cancer. *Cancer Immunol. Immunother.* **67**, 13–23 (2018).

Acknowledgements
We gratefully acknowledge the METABRIC (http://www.cbioportal.org/), TCGA (https://portal.gdc.cancer.gov/) and GEO (https://www.ncbi.nlm.nih.gov/geo/) project organizers as well as all study participants for making the data and results available. Thanks to all the peer reviewers and editors for their opinions and suggestions.

Author contributions
JL (Junyi Li), SL, DL, JM and JL (Jun Li) participated in the design of this study, DZ, YZ (Yibing Zhu), and XX performed the statistical analysis, YZ (Yong Zhang), YW and DZ performed the validation of the signature expression, JL (Junyi Li), SL and YZ (Yibing Zhu) drafted the manuscript, DZ, XX, DL, JM and JL (Jun Li) revised the manuscript. All authors read and approved the final manuscript.

Funding
This work was supported by Scientific and technological innovation project of China Academy of Chinese Medical Sciences (CI2021A01903).

Declarations

Competing interests
The authors declare no competing interests.

Ethics approval
This study did not involve the use of animal and human samples as subjects, and the patient information and expression data used were based on open-source data such as METABRIC, TCGA and GEO, without ethical issues or other conflicts of interest.

Additional information
Supplementary Information The online version contains supplementary material available at https://doi.org/10.1038/s41598-025-03423-8.

Correspondence and requests for materials should be addressed to D.L. or J.L.